Vol 5 — Quantum Mechanics from D_IV^5 (Pedagogical Bridge)

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, pedagogical bridge from substrate to standard QM) + Elie (computational verification)

Reader target

Undergraduate + early graduate physics. Reads Vol 0 (Substrate Foundation) as prerequisite; recovers standard QM (Griffiths / Shankar / Sakurai level) as the substrate-level emergent limit at scales above substrate-tick. Pedagogical bridge volume — connects BST substrate framework to physics-community-standard QM curriculum.

Prerequisites

- Vol 0 Substrate Foundation (D_IV^5 + five integers)
- Vol 1 Ch 2 + Ch 6 + Ch 7 Substrate Hilbert Space + Operator Zoo + Dynamics (basic prerequisite)
- Standard undergraduate QM background (helpful, not required)

BST integer focus

Vol 5 is the substrate → standard QM emergence pedagogical bridge. Demonstrates: - Standard Hilbert space $L^2(\mathbb{R}^3)$ emerges as macroscopic limit of Bergman $H^2(D\_IV^5)$ at coarse-grained scale - Position + momentum + angular momentum + spin standard QM operators emerge from substrate-coset-Cartan-decomposition operators (Vol 1 Ch 6 6/6 zoo) - Schrödinger equation $i\hbar\ \partial|\psi\rangle/\partial t = H|\psi\rangle$ emerges from substrate dynamics H_sub = Casimir on $L^2(D\_IV^5;\ L\_\lambda)$ (Vol 1 Ch 7) at macroscopic scale - Born rule $|\psi|^2$ = Bergman reproducing-kernel evaluation (K67 RATIFIED Tuesday) - Measurement formalism + POVMs derive from substrate observer-bandwidth-bounded recording (T2469 SCMP framework)

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Reader takeaway	Cross-link
1	From Substrate to Standard Hilbert Space	$L^2(\mathbb{R}^3)$ emerges from Bergman H^2(D_IV^5) at macroscopic limit	Vol 1 Ch 2
2	Position + Momentum + Heisenberg	Standard QM Heisenberg algebra emerges from coset Cartan decomposition	Vol 0 Ch 4 §4.2 v0.5; Vol 1 Ch 6
3	Angular Momentum + Spin	Standard SO(3) angular momentum + half-integer spin from substrate SO(5) × SO(2) reduction	Vol 1 Ch 6 + Paper #133 spin-statistics
4	Schrödinger Equation	Standard time-dependent + independent Schrödinger from substrate H_sub dynamics	Vol 1 Ch 7 dynamics
5	Heisenberg Picture + Path Integral	Standard Heisenberg + Feynman path integral as substrate dynamics representations	Vol 1 Ch 7 + Ch 9
6	Hydrogen Atom + Atomic Spectra	Standard hydrogen energy levels + selection rules from substrate-natural Casimir spectrum	Vol 1 Ch 5 Casimir
7	Born Rule + Measurement	Born = Bergman (K67 RATIFIED); POVMs + projective measurement	T2469 SCMP framework
8	Bell-CHSH + Quantum Correlations	Standard Bell experiments + Tsirelson bound + BST sub-Tsirelson 1/8 falsifier	T2399 + Calibration #17
9	Identical Particles + Spin-Statistics	Pauli exclusion + boson/fermion partition from Pin(2) Z_2 grading	Paper #133 v0.2
10	Decoherence + Classical Limit	Macroscopic classical emergence from substrate ensemble-marginalization	T2469 SCMP + Vol 4 Ch 3 boundary
11	POVMs + Quantum Information Basics	Born = Bergman extension to general POVMs	SP-31 #283

12	Pedagogical Bridge Synthesis	Standard QM curriculum recovery + cross-curriculum reading guide	Vol 1 Ch 2-9 + Vol 0 Ch 4-7 cross-link

Approximately 35% existing BST coverage (Vol 1 + Vol 0 substrate framework provides scaffolding); 65% to build at pedagogical-bridge accessibility.

Cross-volume reference map

Inputs from Vol 0: - D_IV^5 + five integers + Bergman H^2(D_IV^5) - Operator zoo 12/14 STRUCTURALLY VERIFIED (T2419, T2422, T2425, T2421, T2399, H_sub Elie + T2470/T2471/T2472) - Substrate-coset Cartan decomposition + Pin(2) Z_2 grading

Inputs from Vol 1: - Bergman Hilbert space + Casimir algebra (Ch 2 + Ch 5) - Schrödinger / Heisenberg / path integral framework (Ch 7) - Operator zoo 6/6 framework-complete (Ch 6) - Substrate-tick UV-completeness (Ch 10)

Outputs to: - Vol 7 Information Theory (Bell-CHSH + Born rule + POVMs) - Vol 8 Biology (substrate-coupled observer framework) - Vol 9 Condensed Matter (many-body QM emergent from substrate)

— Vol 5 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold